

Table 6. Robustness and falsification summary

Test	Result	Implication
Leave-one-ETF-out	External-financing activation remains negative if	Not driven by one fund
Architecture assignment permutation	External finance two-sided $p = 0.0148$; growth-c	Cross-sectional alignment is non-random
Alternative definitions	External finance expected-sign rate = 0.667; co	Sign stability across measurement choices
Sorted portfolios	External finance HML = -0.01017 ($p = 0.0198$);	Portfolio-level economic relevance
Factor-beta controls	Signs persist but precision weakens for external	Part of activation overlaps with conventional style
Future-CPU lead placebo	No lead survives BH adjustment	No support for systematic pre-trend; causal claims
Downside and quantile models	Supporting signs; bootstrap intervals often wide	Secondary, not anchor evidence

Note. HML denotes the high-minus-low architecture portfolio. Multiple-testing adjustments are applied within the pre-specified test families.